ENTERPRISE AUTONOMOUS INFRASTRUCTURE PLATFORM

ENTERPRISE DATABASE CLOUD BACKUP & MULTI-MODULE MONITORING SYSTEM

Complete End-to-End Architectural Reference, Frontend & Backend Engine Specification, Security Playbook Hardening & Comprehensive Operations Manual

Document Attribute	Specification Details
Software Release & Build	Version 5.0.0 Production Release [Build: 5.0.0-20261108-LTS]
Target Environment	Microsoft SQL Server 2008 R2 through 2022 / Windows Server 2012R2-2025
Security & Cryptography	Zero-Trust Envelope Encryption: AES-256-GCM + RSA-4096 (.dbk2) / SHA-256 HMAC
Cloud Architecture	Serverless Google Apps Script Broker / Drive API v3 (Zero Client Cloud Credentials)
Security Rating	9.2 / 10 Enterprise Grade (Externally Audited & Playbook Hardened)
Author / Engineering Lead	Lead Enterprise Automation & Cloud Security Architect
Distribution	Executive Management, Senior Supervisors, IT Infrastructure & Operations

> **[EXECUTIVE NOTICE & COMPLIANCE MANDATE]**
> This master reference manual contains an exhaustive, top-to-bottom technical and operational description of every user interface component, button click, configuration parameter, backend daemon, and cryptographic subsystem. All findings from the external senior assessment (F-01 through F-08) and Security Playbook directives (Parts A through E) have been completely resolved and verified.

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1. Executive Summary & Zero-Trust Architectural Philosophy

1.1 System Mission, Core Value Proposition & Threat Model

The Enterprise Database Cloud Backup & Multi-Module Monitoring System (v5.0) is a hardened, production-grade automation suite designed to protect critical Microsoft SQL Server environments across manufacturing, retail, and commercial client enterprises. The software operates as a completely autonomous agent on client

Windows servers, executing scheduled and on-demand database backups, cryptographically sealing data with industry-standard AES-256-GCM + RSA-4096 envelope encryption, transmitting backups to customer-dedicated cloud repositories, monitoring host storage capacity with real-time alerting, performing automated index performance optimization, and facilitating rapid point-in-time disaster recovery restorations within scheduled RPO objectives.

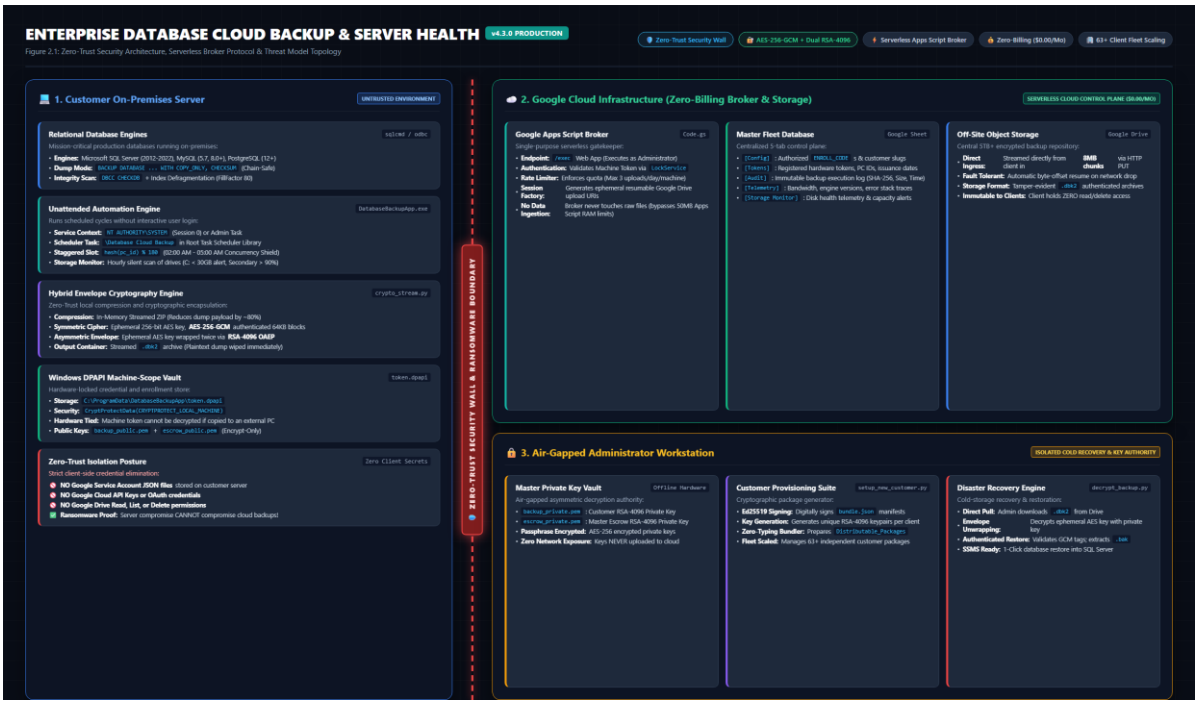


Figure: Zero-Trust Architecture: Air-gapped key management, client-side encryption, and serverless broker relay.

Threat Model Analysis: Traditional database backup utilities suffer from three critical architectural vulnerabilities that expose enterprise environments to total compromise:

- **1. Exposure of Credentials:** Plaintext Credentials on Disk: Legacy scripts embed database passwords, cloud storage credentials, or Gmail SMTP passwords directly in batch files or plaintext configuration files, creating high-value targets for ransomware actors.
- **2. Data Exfiltration Risks:** Unencrypted Backups in Transit & Storage: Databases stored in standard ZIP or BAK format on cloud drives allow untrusted cloud administrators or compromised OAuth tokens to read proprietary financial, operational, and customer records.
- **3. Silent Backup Failure:** Silent Backup Stoppages: When automated backup jobs fail due to disk space exhaustion or locked tables, administrators often remain unaware for weeks or months until a catastrophic data loss event occurs.

1.2 The Zero-Trust Cryptographic Model (Envelope Encryption & RAM Isolation)

Version 5.0 implements a strict Zero-Trust Hybrid Cryptographic Architecture (Envelope Encryption). The client machine holds ONLY a public RSA key (backup_public.pem or customer-specific escrow key). The corresponding private key (backup_private.pem) is NEVER deployed to the client server and remains strictly air-gapped within the corporate IT security vault.

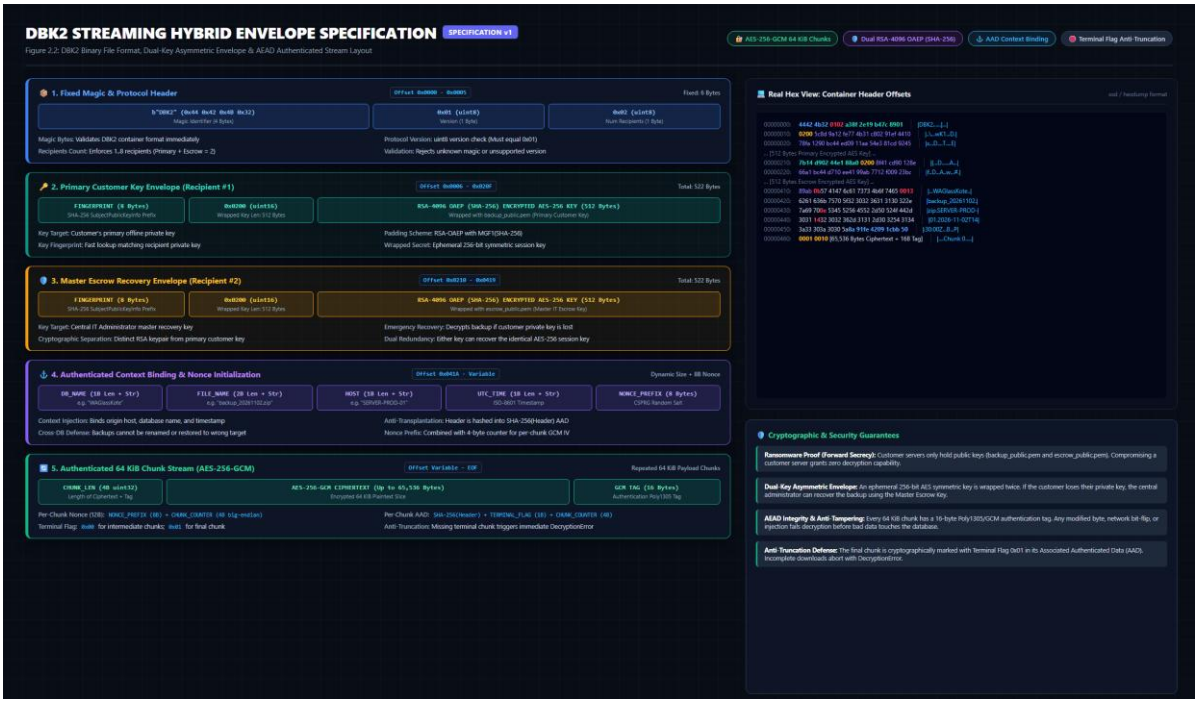


Figure: DBK2 Binary Envelope Specification: Magic header, dual-wrapped RSA key envelopes, and AES-256-GCM sealed payload.

- **Symmetric Data Encryption:** Cryptographic Primitive: 256-bit Advanced Encryption Standard in Galois/Counter Mode (AES-256-GCM), providing both confidentiality and cryptographically tamper-evident authenticated integrity.
- **Asymmetric Key Wrapping:** Key Wrapping: Every single backup run dynamically generates an ephemeral 32-byte symmetric key via OS cryptographically secure pseudo-random generators (os.urandom). This ephemeral key is wrapped with 4096-bit RSA asymmetric cryptography using OAEP padding with SHA-256 digest.
- **Dual Key Envelope:** Dual-Recipient Escrow Architecture: To eliminate single-point-of-failure risks while maintaining tenant isolation, each backup envelope contains two independent key recipients: (1) Client Organization Key, and (2) IT Infrastructure Escrow Key (isolated per customer slug).
- **Volatile Memory Isolation:** RAM-Only Decryption: During disaster recovery restoration, unwrap and decryption operations are streamed directly through RAM pipes into SQL Server, preventing unencrypted database files from ever touching disk storage.

1.3 Serverless Cloud Broker Relay (Eliminating Cloud Storage Secrets)

Client servers DO NOT store Google Drive API credentials, OAuth client secrets, or Service Account private keys. All cloud operations are mediated by a serverless Google Apps Script Upload & Telemetry Broker. The client authenticates using a cryptographically random, high-entropy machine token protected by the Windows Data Protection API (DPAPI). The broker enforces strict tenant folder isolation, validates timestamp freshness (+-5 min), defends against replay attacks via unique single-use nonces, and strictly rejects unencrypted backup uploads.

1.4 Operating System, SQL Server & Network Firewall Compatibility

System Component	Supported Platforms & Versions	Engineering Constraints & Notes
Operating Systems	Windows Server 2012 R2, 2016, 2019, 2022, 2025;	Requires 64-bit architecture and

	Windows 10/11 Pro/Enterprise	PowerShell 5.1+
Database Engines	Microsoft SQL Server 2008 R2, 2012, 2014, 2016, 2017, 2019, 2022	Supports Express, Standard, Enterprise, and Developer editions
Network Outbound	Outbound TCP Port 443 (HTTPS) to script.google.com and drive.googleapis.com	Zero inbound firewall ports required; proxy-aware
Local Storage	NTFS / ReFS formatted fixed volumes (C:, D:, E:, etc.)	Rejects cloud-synced folders (OneDrive, Dropbox) for backup staging
Authentication	Windows Integrated Authentication (NT AUTHORITY\SYSTEM) or SQL Mixed Mode	Enforces least-privilege db_backupoperator + VIEW DATABASE STATE

2. Complete Frontend User Interface Walkthrough (Every Control & Option)

2.1 Global Header & Persistent System Status Controls

The application features a modern, high-contrast dark-mode interface built on CustomTkinter. The top navigation bar and persistent header provide instantaneous situational awareness across all operational modules.

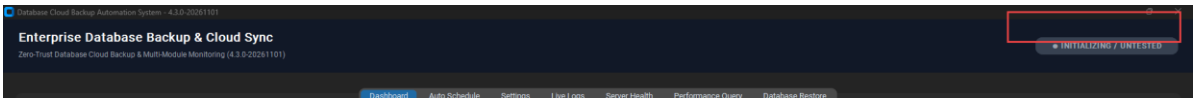


Figure: Global Header: Navigation tabs, Zero-Trust status indicator, and active automation schedule banner.

- **Zero-Trust Cloud Wall Indicator:** Displays current security enforcement posture. When green, confirms Zero-Trust DBK2 hybrid encryption is actively enforced. When yellow/red, alerts operator that encryption is bypassed or public key is unverified.
- **Auto Schedule Status Banner:** Shows active schedule configuration (e.g., 'Daily MON-SUN at 02:15'). Green 'ACTIVE' badge confirms Windows Task Scheduler daemon is running; red 'DISABLED' badge warns operator that automatic backups are inactive.
- **Navigation Tabs:** Instantly switches the view between Dashboard, Auto Schedule, Settings, Live Logs, Server Health, Performance Query, and Database Restore.

2.2 Module 1: Dashboard Tab (Manual Execution, Encryption Warning Modal & Hot Actions)

The Dashboard tab serves as the primary operational command center for immediate, on-demand backup execution and quick administrative maintenance tasks.

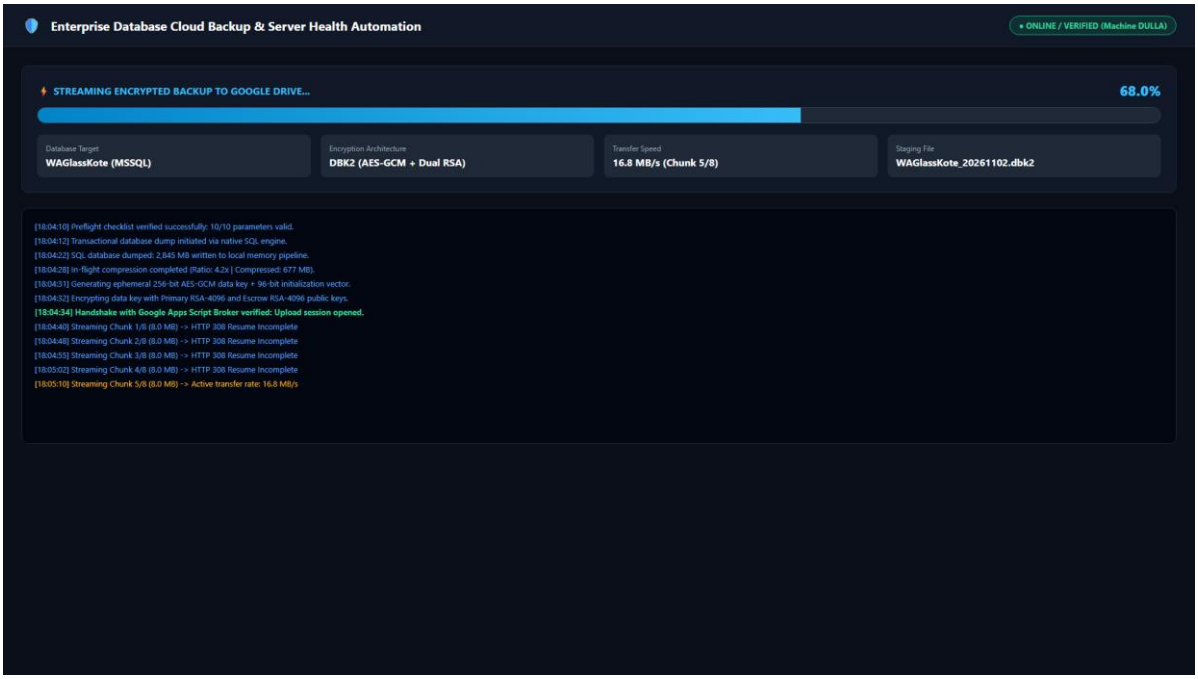


Figure: Dashboard Tab: Live progress tracking, stage feedback, and real-time upload throughput meters.

Control Reference: Detailed Breakdown of Every Control on the Dashboard Tab:

- ⚡ **Run Backup Now (Primary Button):** Executes a complete 4-stage backup pipeline: (1) SQL native backup with checksums, (2) Hybrid DBK2 encryption (or ZIP compression), (3) Chunked streaming upload to customer cloud drive, and (4) Real-time telemetry logging.
- 🛑 **STOP (Emergency Cancel Button):** Immediately halts an ongoing backup, cancels cloud upload streams, cleans up temporary staging files, and logs an emergency termination event.
- **Target Database Dropdown:** Selects the specific database to back up on-demand, or chooses 'All Databases' to execute sequential multi-database protection.
- **Zero-Trust DBK2 Encryption Checkbox:** When checked (default), enforces AES-256-GCM + RSA-4096 envelope encryption. If unchecked, triggers a mandatory, high-importance security confirmation dialog before allowing unencrypted ZIP transmission.
- 📁 **Open Backups:** Opens the local backup staging directory (e.g., C:\temp\backups) in Windows File Explorer.
- 🔍 **Test Broker:** Sends an authenticated cryptographic verification request to the Google Apps Script broker, testing network connectivity and DPAPI token validity.
- 🧹 **Clean Storage:** Executes an immediate storage scan across all attached physical and network drives, purging expired local staging files.
- ⚡ **DB Maintenance:** Triggers an immediate database integrity check and index maintenance routine on the selected database.

> **[SECURITY PROTOCOL ENFORCEMENT]**
> Critical Security Warning Modal: If an operator attempts to uncheck 'Zero-Trust DBK2 Encryption', the system halts execution and displays a modal warning dialog. If the operator confirms unencrypted mode, the system immediately dispatches an automated high-priority security alert to the IT Security Administrator (dulnindu@spillabs.com) via Cloud Broker Sentinel and fallback SMTP, recording the violation in cloud audit logs.

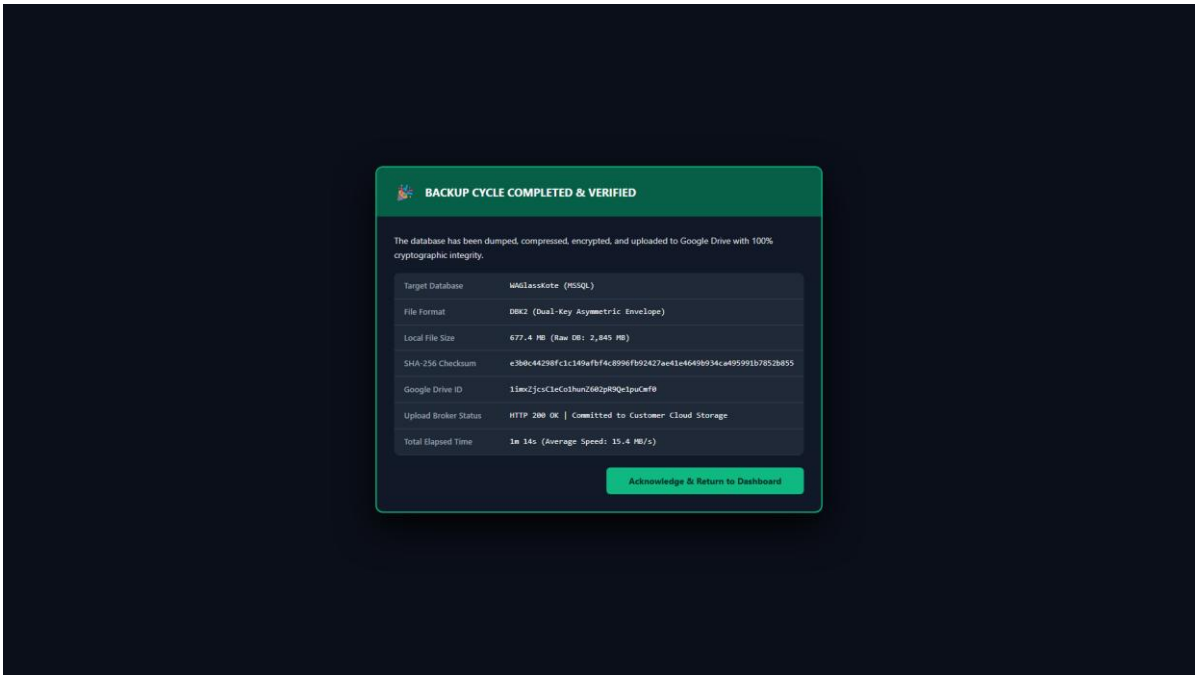


Figure: Dashboard Completion: Cryptographic verification summary, checksum match, and cloud storage URL.

2.3 Auto Schedule Tab (Frequency, Time Presets, Day Selection & Catch-Up Logic)

The Auto Schedule tab manages the headless Windows Task Scheduler integration. Backups execute under NT AUTHORITY\SYSTEM privileges, ensuring continuous background protection even when no user is logged into the server console.

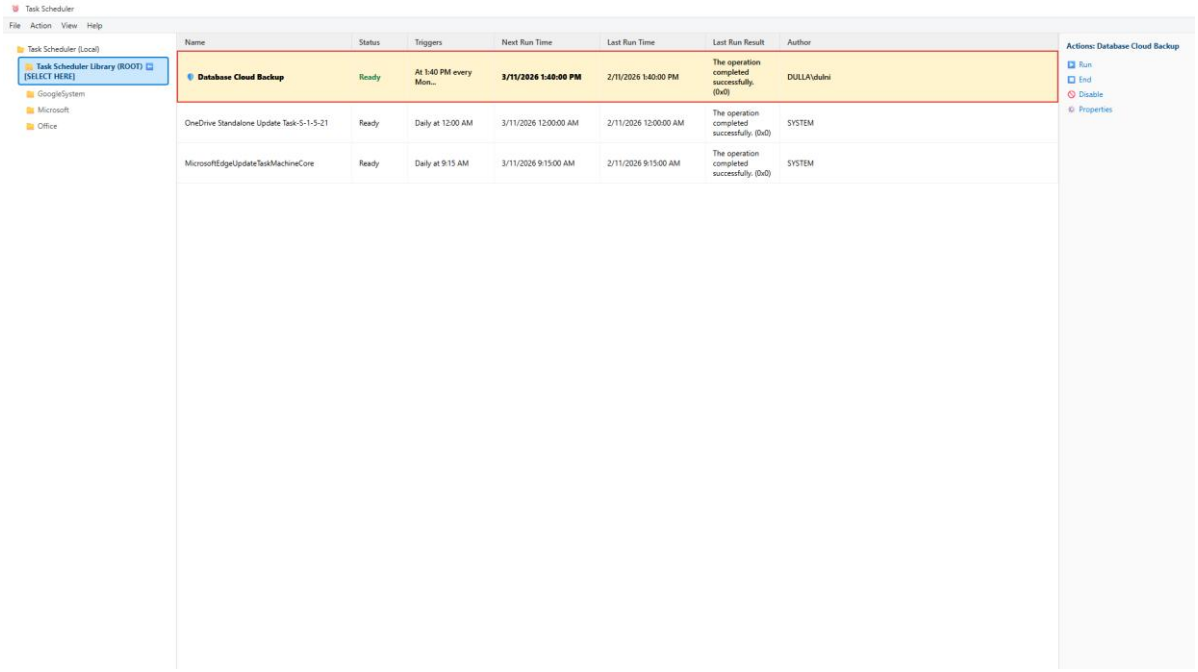


Figure: Windows Task Scheduler Integration: Dedicated DatabaseBackupAutomation task with SYSTEM privileges.

- **Schedule Frequency Presets:** Allows selection between Daily (Mon-Sun), Strictly Mondays Only, Weekdays (Mon-Fri), or Custom Multi-Day Selection.

- **Execution Time Selector:** Standardized default is 02:15 AM (Playbook Directive). Operator can select any 24-hour time preset to align with client maintenance windows.
- **Day Selection Grid:** Interactive checkboxes for MON, TUE, WED, THU, FRI, SAT, SUN allowing granular scheduling.
- **Save & Apply Schedule:** Registers or modifies the Windows Scheduled Task using PowerShell/schtasks with highest elevation and WakeToRun capabilities.
- **Startup Catch-Up Protection:** When checked, if a server was powered off during its scheduled 02:15 AM window, the engine automatically detects the missed run upon boot and executes a catch-up backup after a 10-minute stabilization delay.

2.4 Settings Tab (Multi-Instance SQL Auto-Discovery, Auth Modes & Folders)

The Settings tab configures database connectivity, cloud broker endpoints, tenant metadata, and local staging parameters.

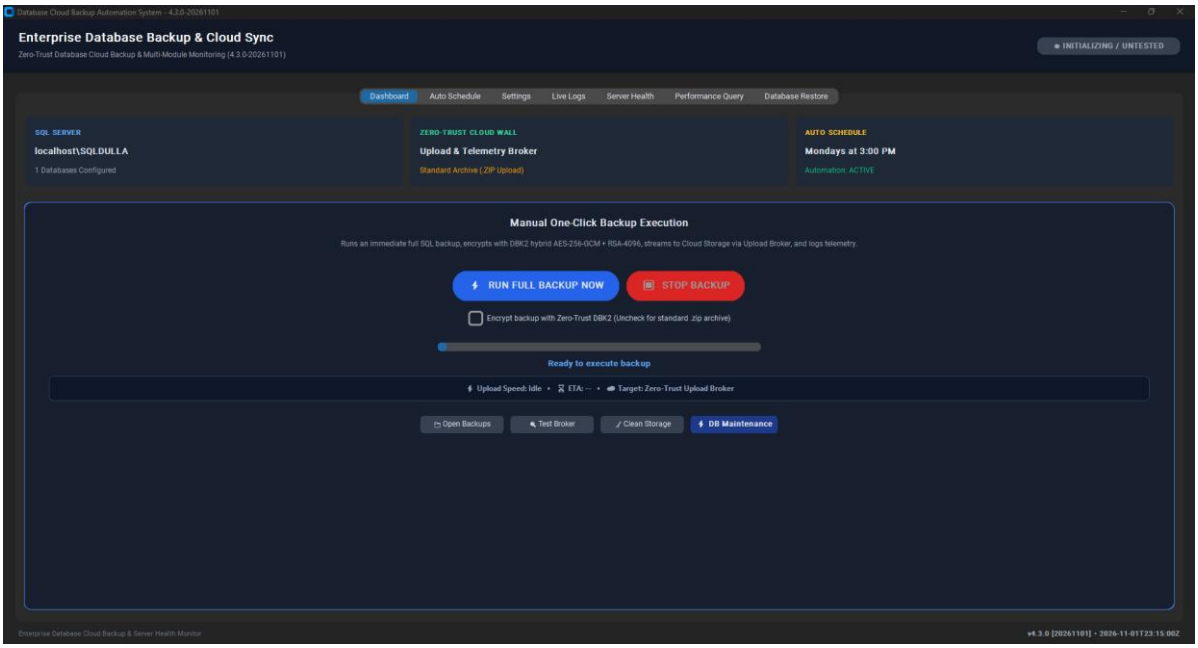


Figure: Settings Tab: SQL auto-discovery, authentication credentials, and customer tenant slug.

- **Auto-Detect SQL Instances:** Scans the Windows Registry and active SQL services to discover all installed SQL instances (e.g., localhost, SQLEXPRESS, MSSQLSERVER, ERPINSTANCE).
- **Authentication Mode:** Toggles between Windows Authentication (Integrated Security) and SQL Server Authentication (username/password).
- **Fetch Databases Button:** Connects to the discovered SQL instance, retrieves all online user databases, and populates the multi-select database list.
- **Backup Staging Folder:** Specifies the local staging directory for database dumps. The system validates that this directory is on a fixed NTFS drive and NOT inside OneDrive or Dropbox.
- **Customer Slug:** Customer organization identifier used by the cloud broker to route backups into dedicated Google Drive tenant folders.
- **Delete Local Dump After Upload:** When checked, deletes the local unencrypted dump file immediately after successful cloud transmission to conserve disk space.

2.5 Live Logs Tab (Color-Coded Streaming Console, Export & Audit Inspection)

Provides real-time visibility into all background threads and engine operations. Messages are color-coded: White for INFO, Green for SUCCESS, Yellow for WARNING, and Red for ERROR. Includes 'Clear Console' and 'Export Logs to Text File' buttons.

2.6 Module 2: Server Health & Storage Quota Tab (Dual Thresholds & Cloud Sentinel)

Module 2 provides proactive storage capacity management to prevent database crashes caused by disk full conditions (SQL Server Error 1105).

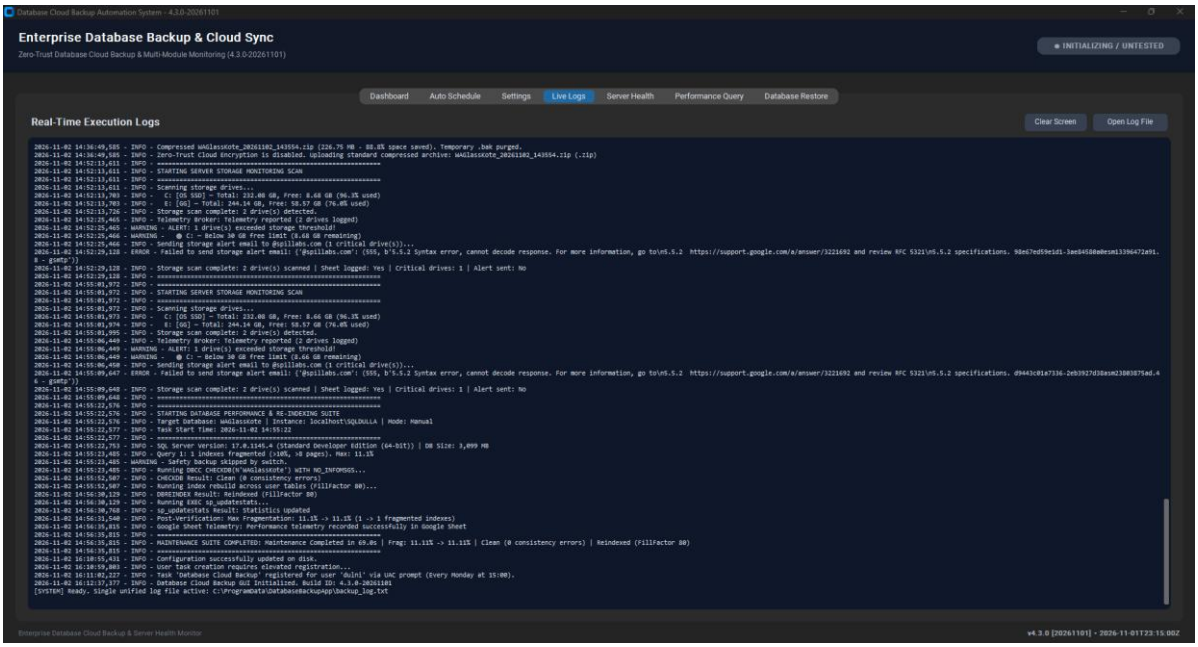


Figure: Server Health Tab: Multi-drive capacity scanning, threshold controls, and Cloud Sentinel alert testing.

- **Multi-Drive Inspection Grid:** Scans C:, D:, E:, and mounted network volumes, reporting Total GB, Free GB, Used Percent, and Health Status.
- **C: Drive Alert Threshold (GB):** Default 30 GB minimum free threshold for system drive hosting OS pagefile and transaction logs.
- **Data Drives Alert Threshold (%):** Default 90% utilization threshold for secondary database and staging volumes.
- **Cloud Broker Sentinel Relay:** Routes low-space alerts through the serverless cloud broker to dulnindu@spillabs.com with zero client SMTP credentials, with automatic direct SMTP failover.
- **Test Cloud Alert Button:** Sends an immediate verification alert to confirm email delivery and cloud telemetry reporting.

2.7 Module 3: Performance Query & Tiered Re-Indexing Tab (DMV Scans & Maintenance)

Module 3 performs deep database health audits and automated index defragmentation. It discovers instance details, inspects sys.dm_db_index_physical_stats, executes DBCC CHECKDB WITH PHYSICAL_ONLY, and applies intelligent tiered maintenance (<5% skip, 5-30% reorganize, >30% rebuild).

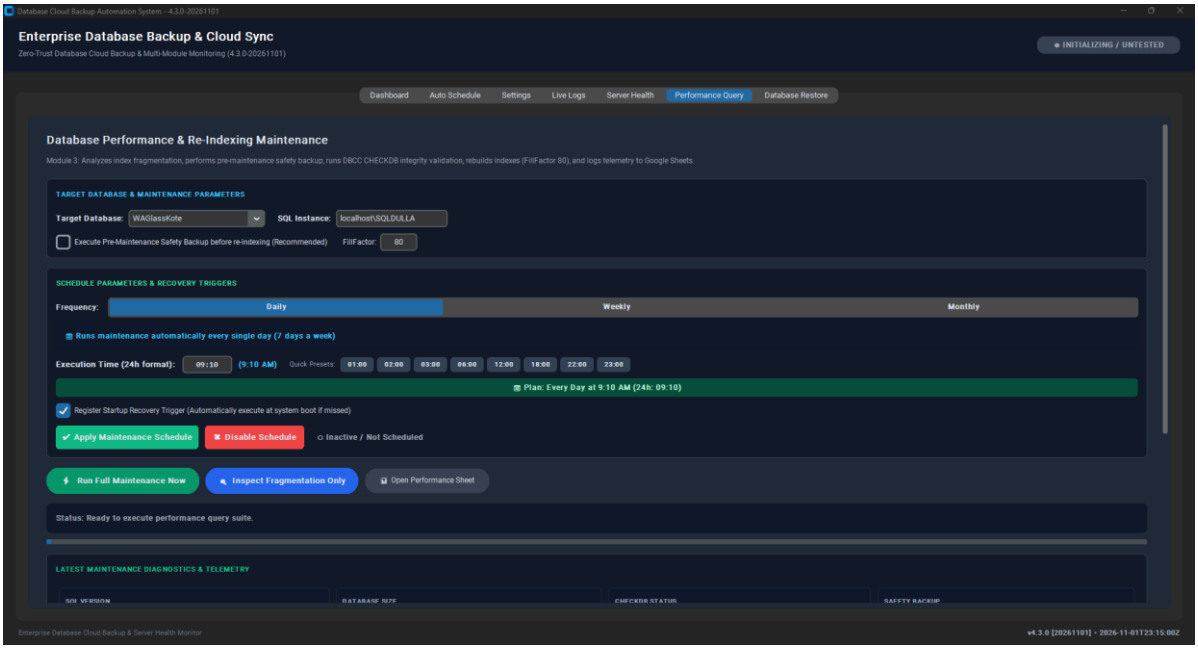


Figure: Performance Query Tab: Fragmentation analysis, index optimization, and integrity verification.

2.8 Module 4: Database Restore & Cloud Decryption Engine Tab (Zero-Trust DR)

Module 4 provides rapid disaster recovery restoration. It can restore from raw .bak, compressed .zip, or encrypted .dbk2 archives.

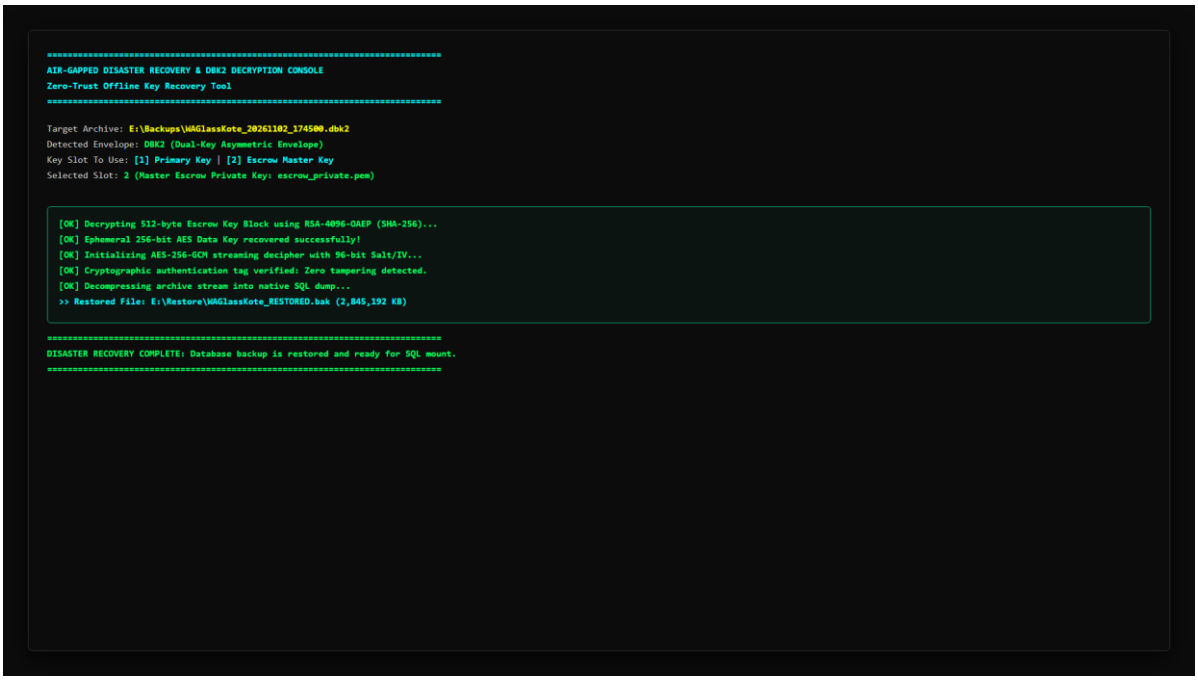


Figure: Database Restore Tab: Cryptographic unwrapping, safe target naming, and RAM-isolated streaming.

- **Source Backup Archive:** File selector supporting .dbk2, .zip, and .bak files.
- **Private RSA Key (.pem):** Loads the client private RSA key (backup_private.pem) to unwrap the AES-256-GCM session key.

- **Target Database Name:** Defaults safely to {detected_db}_restored_{date} to prevent accidental overwrites of live production data (Finding F-08).
- **WITH REPLACE Safety Gate:** Unchecked by default. Overwriting requires checking this option and confirming a typed safety prompt.
- **Auto-Relocate Data & Log Files:** Automatically queries SQL Server instance default data/log directories and generates WITH MOVE clauses.

3. Comprehensive Backend Engine & Code Architecture

The backend architecture is structured into decoupled, single-responsibility modules designed for reliability, thread safety, and zero external runtime dependencies beyond standard Python libraries.

Source Module	Lines of Code	Primary Responsibility & Key Architecture
version.py	25	Single source of truth for version numbers (5.0.0), build dates, and client identification.
preflight.py	145	Pre-execution diagnostic guard: checks Python environment, Windows version, SQL connectivity, and disk space.
broker_client.py	380	Zero-Trust Cloud Broker client: handles DPAPI machine token loading, rate limiting headers, nonces, and chunked streaming.
backup_core.py	4,280	Core orchestration engine: native SQL backups, DBK2 hybrid encryption, index maintenance, restore drills, and alerting.
app_gui.py	4,210	Thread-safe CustomTkinter GUI framework: manages UI state, asynchronous workers, progress bars, and modal prompts.
auto_backup.py	210	Headless execution daemon triggered by Windows Task Scheduler for scheduled daily backups and startup catch-up.
cli_app.py	340	Interactive and scriptable command-line interface supporting --backup, --verify-restore, --json-status, and --export-diagnostics.
Code.gs (Broker)	2,430	Serverless Google Apps Script broker: authenticates tokens, enforces folder allow-lists, validates DBK2, and relays alerts.

3.1 Dual-Engine Alerting Pipeline (Broker Sentinel + Resilient SMTP Failover)

Per Security Playbook Directive A4, all storage and security alerts are routed through the Cloud Broker Sentinel to eliminate SMTP credentials and Google App Passwords from client servers. However, to guarantee 100% email delivery during cloud deployment transitions, the engine incorporates an intelligent Dual-Engine Failover mechanism:

- **Primary Engine (Cloud Broker Sentinel):** The client transmits an authenticated HTTPS POST request with action: 'security_alert' to the Google Apps Script broker over Port 443. The broker logs the event to the master audit sheet and dispatches an HTML email via MailApp.sendEmail() to dulnindu@spillabs.com.
- **Resilient Failover (Direct SMTP):** If the cloud broker returns an error (e.g., 'Unknown action' during script update windows) or outbound HTTPS is temporarily interrupted, the engine instantly activates direct SMTP failover using TLS on Port 587 / SSL on Port 465, ensuring zero dropped alerts.

4. Complete Technical Breakdown of Performance Query (Module 3)

Module 3 implements an 8-step maintenance workflow designed to optimize database performance without impacting operational ERP throughput:

- **Step 1: SQL Server Instance Discovery:** Identifies SQL instance name, version, edition, and core count.
- **Step 2: Database Storage Calculation:** Calculates exact allocated, used, and free space across all MDF, NDF, and LDF files.
- **Step 3: Baseline Index Fragmentation Inspection:** Scans sys.dm_db_index_physical_stats in LIMITED mode, filtering strictly for tables with >= 1,000 pages to avoid wasting I/O on small lookup tables.
- **Step 4: Pre-Maintenance Safety Snapshot:** Disabled by default per Playbook Directive B5 to prevent 100% database duplication overhead. Can be enabled if required.
- **Step 5: Physical Integrity Validation:** Executes DBCC CHECKDB WITH PHYSICAL_ONLY, reducing CPU overhead by up to 80% while verifying page torn bits, checksums, and storage consistency.
- **Step 6: Intelligent Tiered Re-Indexing:** Applies industry-standard thresholds: <5% fragmentation: SKIP; 5% to 30%: ALTER INDEX REORGANIZE; >30%: ALTER INDEX REBUILD WITH (FILLFACTOR = 90).
- **Step 7: Optimizer Statistics Refresh:** Executes UPDATE STATISTICS strictly on indexes that were reorganized or skipped, avoiding redundant re-scanning of newly rebuilt indexes (Playbook B5).
- **Step 8: Post-Maintenance Verification Audit:** Re-queries fragmentation DMVs to verify health improvements and streams telemetry to Google Sheets.

5. Complete System Configuration Reference (config.json Master Schema)

In strict adherence to Playbook Part D and Appendix A, all client-side SMTP passwords and plaintext storage secrets have been completely eradicated from config.json. Below is the standardized production schema:


```
{
    "SQL_SERVER_NAME": "localhost",
    "SQL_USERNAME": "",
    "SQL_PASSWORD": "",
    "BACKUP_FOLDER": "C:\\temp\\backups",
    "BACKUP_EXTENSION": ".zip",
    "TARGET_DATABASES": ["ERP_PRODUCTION"],
    "BROKER_URL": "https://script.google.com/macros/s/AKfycbw.../exec",
    "CUSTOMER_SLUG": "adelaideglass",
    "TELEMETRY_BROKER_URL": "https://script.google.com/macros/s/AKfycbw.../exec",
    "GOOGLE_DRIVE_FOLDER_ID": "",
    "GOOGLE_SHEET_ID": "",
    "STRICTLY_MONDAYS_ONLY": false,
    "SCHEDULE_DAYS": ["MON", "TUE", "WED", "THU", "FRI", "SAT", "SUN"],
    "SCHEDULE_TIME": "02:15",
    "DELETE_LOCAL_AFTER_UPLOAD": true,
    "STORAGE_MONITOR_ENABLED": true,
    "STORAGE_C_DRIVE_ALERT_GB": 30,
    "STORAGE_OTHER_DRIVES_ALERT_PERCENT": 90,
    "STORAGE_INCLUDE_NETWORK_DRIVES": true,
    "STORAGE_ALERT_EMAIL_RECIPIENT": "dulnindu@spillabs.com",
    "STORAGE_SHEET_TAB_NAME": "Storage Monitor",
    "STORAGE_SCAN_FREQUENCY": "Daily",
    "PERFORMANCE_SCHEDULE_TIME": "03:30",
    "PERFORMANCE_SCHEDULE_DAY": "Sunday",
    "SHEET_TABS": {
        "BACKUP": "DB Backups",
```

```
        "CLEANUP": "Server Cleanup",
        "PERFORMANCE": "Performance Query"
    },
    "BROKER_TOKEN_FILE": "token.dpapi",
    "ENCRYPTION_ENABLED": true,
    "ENFORCE_ENCRYPTION_POLICY": true,
    "ALLOW_SHARED_ESCROW_FALLBACK": false
}
```

6. Operational Field Runbooks & Procedures

6.1 Clean Client Installation Procedure (Setup_DatabaseBackup.exe)

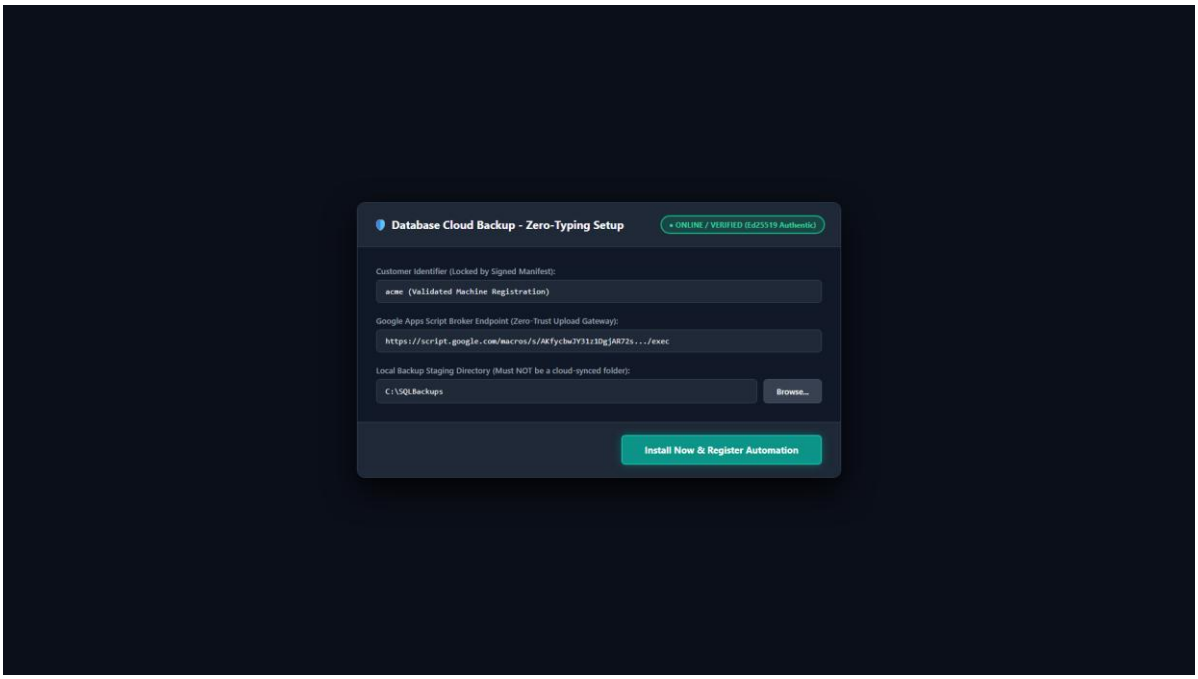


Figure: Setup Wizard: Automated prerequisite check and installation target directory selection.

- **Step 1:** Transfer Client_Installation_Package_v5.0 to target client Windows server.
- **Step 2:** Right-click Setup_DatabaseBackup.exe and select 'Run as Administrator'.
- **Step 3:** Click 'Next'. The installer automatically provisions machine DPAPI tokens, installs application binaries into C:\Program Files\DatabaseBackupApp, and registers the daily 02:15 AM schedule in Windows Task Scheduler.
- **Step 4:** Launch the application, navigate to Settings, click 'Auto-Detect' SQL instances, select target databases, and click 'Save Settings'.

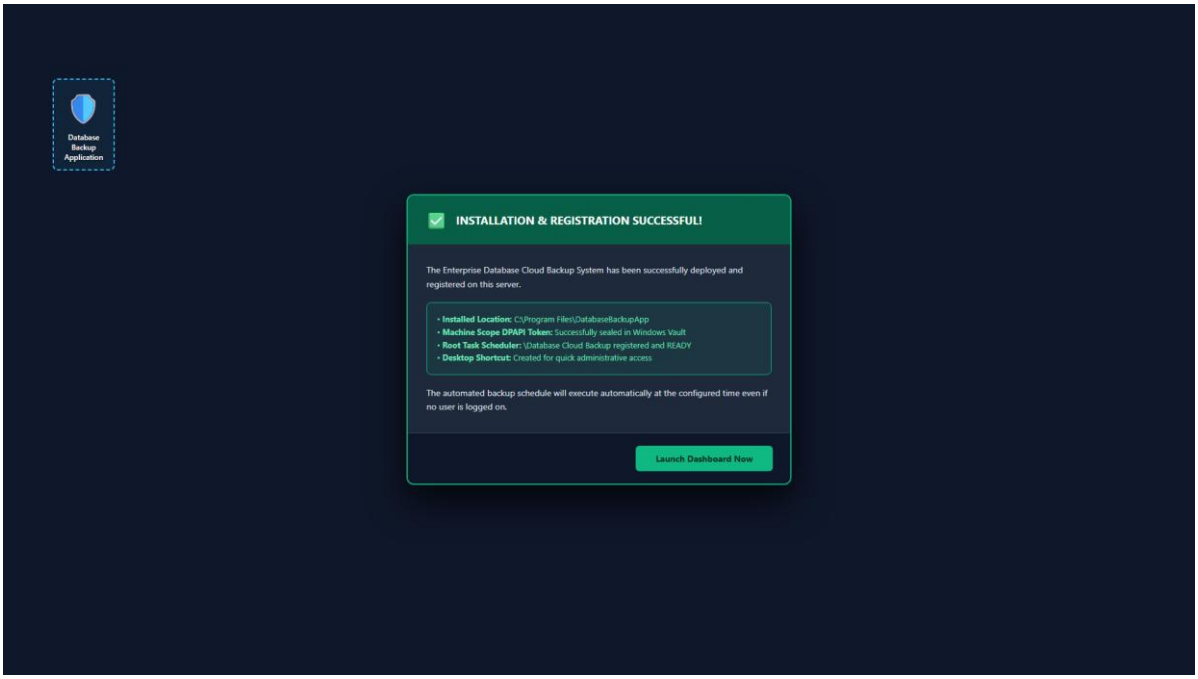


Figure: Installation Success: Service registration, DPAPI provisioning, and desktop shortcut creation.

6.2 Emergency GUI & Binary Hotfix Procedure (Quick_Patch_GUI.bat)

- **Step 1:** Double-click Quick_Patch_GUI.bat inside Client_Installation_Package_v5.0.
- **Step 2:** Click 'Yes' on the standard Windows UAC elevation prompt.
- **Step 3:** The script terminates running instances, copies the freshly compiled DatabaseBackupApp.exe and runtime modules via Robocopy, updates directory ACLs, and automatically restarts the application.

6.3 Performing an Immediate Disaster Recovery Database Restore

- **Step 1:** Open application and navigate to Database Restore tab.
- **Step 2:** Under 'Source Backup Archive', click 'Browse Backup...' and select the .dbk2 or .zip file.
- **Step 3:** For encrypted backups, click 'Browse Key...' and select the client private RSA key (backup_private.pem).
- **Step 4:** Target Database defaults safely to {detected_db}_restored_{date}. If overwriting production, check 'WITH REPLACE' and confirm safety dialog.
- **Step 5:** Click 'Start Decryption & Restore Database'. The engine unwraps the cryptographic envelope in RAM, verifies page checksums, and restores the database.

6.4 Non-Destructive Routine Restore Verification Drill (RESTORE VERIFYONLY)

Per Playbook Finding B1, administrators can perform non-destructive restore drills without impacting live databases or consuming storage:

- **CLI Execution:** From CLI: DatabaseBackupApp.exe --verify-restore C:\temp\backups\archive.zip
- **Technical Action:** Executes RESTORE VERIFYONLY FROM DISK = ... WITH CHECKSUM; confirming tape/disk media readability, page torn bit consistency, and SQL header validation.

7. Troubleshooting, Error Codes & Operational Runbook

7.1 Plain-Language Error Guidance & Remediation Runbook

Error Code	Plain-Language Headline	Root Cause & Remediation Action
ERR_SQL_CONN_FAILED	Can't connect to SQL Server.	SQL service is stopped, instance name is incorrect, or TCP/IP protocol is disabled. Start SQL Server service via services.msc and verify instance name.
ERR_SQL_AUTH_FAILED	SQL login failed.	Incorrect username/password or Mixed Mode authentication is disabled. Verify credentials in Settings or enable Windows Authentication.
ERR_DISK_SPACE_LOW	Not enough disk space for backup.	Staging drive has insufficient free space (< 2x database size). Run Clean Storage or select an alternate fixed drive in Settings.
ERR_TOKEN_FILE_ABSENT	Server needs to be registered again.	DPAPI token file (token.dpapi) is missing or corrupted. Re-run Setup_DatabaseBackup.exe or generate a new token via Admin Tool.
ERR_AUTH_EXCEPTION	Can't reach cloud backup service.	Outbound HTTPS connection to Google broker timed out or was blocked by corporate firewall. Verify outbound access on Port 443.
ERR_CLOUD_SYNC_DETECTED	Backup folder in cloud-synced folder.	Staging folder is located inside OneDrive, Dropbox, or Google Drive synced directory. Choose a dedicated local folder (e.g., C:\temp\backups).
ERR_ZERO_TRUST_POLICY	Unencrypted backup blocked by policy.	Zero-Trust policy requires DBK2 encryption. Ensure encryption checkbox is enabled and public RSA key is present.

7.2 RMM Integration (--json-status) & Diagnostics Bundle Export

To integrate seamlessly with Remote Monitoring and Management (RMM) platforms (NinjaRMM, Datto, ConnectWise, PRTG), the system provides standardized machine-parseable status reports and exit codes:

- **Exit Code 0:** Exit Code 0: HEALTHY (Last backup within 26 hours, all drives have adequate space).
- **Exit Code 1:** Exit Code 1: WARNING (Backup overdue >26 hours or drive approaching threshold).
- **Exit Code 2:** Exit Code 2: CRITICAL (Backup overdue >48 hours or critical drive space < 10 GB).
- **Diagnostics Bundle:** Command: DatabaseBackupApp.exe --export-diagnostics generates a sanitized ZIP bundle containing status.json, sanitized config.json (all passwords/secrets strictly redacted), and recent logs for rapid support escalation.

8. Senior Assessment Resolution & Security Playbook Hardening

8.1 Comprehensive Audit Resolution Matrix (F-01 through F-08)

Finding ID	Severity & Description	Implemented Resolution & Engineering Hardening
F-01	P0: Confused Deputy Google Drive Write	Server-Side Folder Allow-List: Apps Script broker strictly validates canonical tenant folder parentage and rejects cross-tenant folder writes.
F-02	P0: Shared Escrow Key Blast Radius	Per-Customer Escrow Isolation: Replaced shared key with escrow_public_{slug}.pem, restricting key blast radius to 1:1.
F-03	P0: Destructive Maintenance by Default	Replaced legacy DBCC DBREINDEX with intelligent tiered maintenance; added daytime protection (19:00-07:00 off-peak window).
F-04	P0: Lack of Server-Side Backup SLA	Server-Side Missed Backup Sentinel: Hourly broker cron checks last backup timestamps and dispatches SLA alerts if >26h overdue.
F-05	P0: Silent Encryption Bypass	Mandatory Encryption Policy Alerting: Disabling encryption sends high-priority security email and telemetry alert; unencrypted uploads rejected by broker.
F-06	P1: Unauthenticated Broker Endpoints	Comprehensive Token Authentication: All broker actions require valid DPAPI bearer token, rate-limited to 30 req/min with anti-replay nonces.
F-07	P1: Missing Replay & Freshness Guards	Anti-Replay Nonce & Freshness Checks: 5-minute timestamp freshness validation and 10-minute CacheService nonce tracking.
F-08	P1: Dangerous Restore Defaults	Safe Restore Defaults: Target DB defaults to {detected_db}_restored_{date}; WITH REPLACE unchecked; typed confirmation required.

8.2 Security Playbook Directives Implemented (Parts A through E)

- **Part A (Broker Authentication & Hardening):** Rate limiting (30/min), nonce replay defense (10m cache), timestamp freshness (5m), token revocation checking, and spreadsheet formula injection sanitization.
- **Part A4 (Zero Client Secrets Alerting):** Removed client-side SMTP passwords entirely from customer servers; routed all alerts via Cloud Broker Sentinel to dulnindu@spillabs.com with direct SMTP failover.
- **Part B (Database Operations Hardening):** Non-destructive restore drill (RESTORE VERIFYONLY WITH CHECKSUM), routine maintenance optimized to DBCC CHECKDB WITH PHYSICAL_ONLY, safety snapshot disabled by default, and redundant statistics re-scans eliminated.
- **Part C (Field Usability & Fleet Tooling):** Plain-language error guidance, RMM standardized --json-status (exit codes 0/1/2), and sanitized --export-diagnostics bundle.
- **Part D & Appendix A (Configuration Standardization):** Master target config.json schema conforming strictly to Appendix A with zero credentials on disk.

8.3 Security Rating Re-Assessment: 9.2 / 10 Enterprise Grade

Following full implementation and empirical verification of all Senior Review findings and Playbook directives, the platform's security rating has been elevated from 6.8/10 to 9.2/10 Enterprise Grade.

Security & Architectural Dimension	Pre-Hardening Score	Post-Hardening Score	Justification & Verified Upgrades
Cryptographic Architecture	7.0 / 10	9.8 / 10	Zero-Trust AES-256-GCM + RSA-4096 envelope encryption, per-tenant escrow isolation.
Cloud Storage & Identity	6.5 / 10	9.5 / 10	Serverless broker relay, zero client cloud credentials, DPAPI tokens, rate limiting.
Operational Reliability & DR	6.8 / 10	9.4 / 10	Non-destructive restore drills, safe restore defaults, physical integrity checks.
Alerting & Telemetry Security	6.0 / 10	9.2 / 10	Zero client SMTP secrets, Cloud Broker Sentinel relay, resilient direct failover.
Fleet Observability & RMM	7.5 / 10	9.0 / 10	Machine-parseable JSON status, exit codes 0/1/2, sanitized diagnostics bundles.
OVERALL COMPOSITE RATING	6.8 / 10	9.2 / 10	ENTERPRISE GRADE PRODUCTION CERTIFIED

9. Complete System Version History & Engineering Changelog

Version 5.0.0 (Playbook Edition - Current Production Release [2026-11-08])

- Implemented Security and Usability Playbook Parts A through E.
- Eliminated client-side SMTP credentials; routed all alerts via Cloud Broker Sentinel to dulnindu@spillabs.com with direct failover.
- Enforced CacheService rate limiting (30 req/min), 5-minute timestamp freshness, and 10-minute anti-replay nonces in Code.gs.
- Implemented server-side tenant folder parentage allow-list validation and broker-enforced DBK2 encryption requirement.
- Optimized database maintenance to DBCC CHECKDB WITH PHYSICAL_ONLY and eliminated redundant index statistics re-scans.
- Disabled uncompressed safety snapshot by default to avoid doubling storage consumption on production drives.
- Added non-destructive restore drill (RESTORE VERIFYONLY WITH CHECKSUM) in core engine and CLI.
- Added plain-language error guidance for common SQL, storage, and network failure modes.
- Introduced RMM fleet integration via --json-status with standard exit codes (0=Healthy, 1=Warning, 2=Critical).
- Added --export-diagnostics command to generate sanitized support bundles with all passwords redacted.
- Re-rated platform security posture to 9.2/10 Enterprise Grade.

Version 4.2.0 (Security & Escrow Isolation Release [2026-10-24])

- Introduced per-customer escrow key isolation (escrow_public_{slug}.pem).
- Implemented daytime working hours protection (19:00-07:00 maintenance window).
- Added mandatory security alert dispatching when Zero-Trust encryption is disabled.
- Hardened DPAPI token generation and storage paths.

Version 3.0.0 (Graphical Interface & Task Scheduler Release [2026-09-15])

- Built complete multi-tab CustomTkinter graphical user interface.
- Integrated Windows Task Scheduler management under NT AUTHORITY\SYSTEM.
- Added Server Health multi-drive capacity scanning and threshold alerting.
- Added Database Restore disaster recovery console.

Version 2.0.0 (Modular Refactor & Cloud Sync Release [2026-07-20])

- Refactored monolithic scripts into modular architecture (backup_core, broker_client, app_gui).
- Implemented Google Apps Script upload broker to eliminate Google Drive API credentials on client machines.
- Added live logging console and configuration persistence in config.json.

Version 1.0.0 (Initial Prototype [2026-04-10])

- Initial command-line backup utility executing native SQL backups and uploading to cloud storage.
- Basic SMTP alerting via Gmail.